

Problem I

Problem (Problem I.1): Show that in a directed set $(I, \leq)$, for all $n \geq 0$ and for all $i_1, \dots, i_n \in I$, there exists $k \in I$ such that $k \geq i_1, \dots, k \geq i_n$.

Solution: Let $(I, \leq)$ be a directed set, $n \geq 0$, and $i_1, \dots, i_n \in I$.

Let $k_1 = i_1$. Since $(I, \leq)$ is a directed set, there is some $k_2 \in I$ such that $k_1 \leq k_2$ and $i_2 \leq k_2$. By the same reasoning, there is some $k_3 \in I$ such that $k_2 \leq k_3$ and $i_3 \leq k_3$.

This process can be continued n times to find some $k_n \in I$ such that $k_{n-1} \leq k_n$ and $i_n \leq k_n$.

Since $(I, \leq)$ is a preordered set, $\leq$ is transitive, so for any $j \in [n]$, $i_j \leq k_{1,\dots,j} \leq k_{1,\dots,j,j+1} \leq \dots \leq k_{1,\dots,n-1} \leq k_n \Rightarrow i_j \leq k_n$. In other words, there is some $k = k_n \in I$ such that $k \geq i_1, \dots, k \geq i_n$.

Problem (Problem I.2): Let J be a subset of a directed set I . We say J is cofinal if for all $i \in I$, there exists $j \in J$ such that $i \leq j$. We equip J with the induced preorder inherited from I . Show that if J is cofinal, then J is also a directed set.

Solution: Let J be a cofinal subset of a directed set I .

It is clear that $(J, \leq)$ is still a preordered set because if there were any $i, j, k \in J$ which did not satisfy the properties of reflexivity or transitivity, this would contradict that $(I, \leq)$ is a preordered set.

For any $i, j \in J$, we know that $i, j \in I$. Since $(I, \leq)$ is directed, there is then some $m \in I$ such that $i \leq m$ and $j \leq m$. But, J is cofinal, so there must be some $k \in J$ such that $m \leq k$. Then, since $(I, \leq)$ is preordered, we know that $i \leq k$ and $j \leq k$.

Finally, as $I \neq \emptyset$, we can take some $i \in I$. Since J is cofinal, there must be some $j \in J$ such that $i \leq j$, so we know that $J \neq \emptyset$ too.

Thus, $(J, \leq)$ is also a directed set.

Problem (Problem I.3): Show that for a net $(x_i)_{i \in I}$ in a Hausdorff separated space $(X, \mathcal{T}_X)$, if it converges to some element $\bar{x}$, then this element is unique.

Solution: Let $(X, \mathcal{T}_X)$ be a Hausdorff separated space and $(x_i)_{i \in I}$ be a net converging to some $\bar{x} \in X$.

Suppose towards a contradiction that there is some other $\bar{y} \in X$ such that $\bar{x} \neq \bar{y}$ and $x_i \rightarrow \bar{y}$. Then, since $(X, \mathcal{T}_X)$ is Hausdorff, there are some $U, V \in \mathcal{T}_X$ such that $\bar{x} \in U$, $\bar{y} \in V$, and $U \cap V = \emptyset$.

However, as $(x_i)_{i \in I}$ converges to both $\bar{x}$ and $\bar{y}$, there are some $k_{\bar{x}}, k_{\bar{y}} \in I$ such that $i \geq k_{\bar{x}} \Rightarrow x_i \in U$ and $j \geq k_{\bar{y}} \Rightarrow x_j \in V$.

Since I is a directed set, there is some $N \in I$ with $N \geq k_{\bar{x}}$ and $N \geq k_{\bar{y}}$. In other words, we have that $x_N \in U$ and $x_N \in V$, which contradicts that $U \cap V = \emptyset$.

Thus, $\bar{x}$ must be the unique element that $(x_i)_{i \in I}$ converges to.

Problem (Problem I.4): Consider a function $f: X \rightarrow Y$ between topological spaces. Let $\bar{x} \in X$ and $\bar{y} \in Y$. We say $\lim_{x \rightarrow \bar{x}} f(x) = \bar{y}$ if and only if for all $V \in \mathcal{O}_Y(\bar{y})$, there exists $U \in \mathcal{O}_X(\bar{x})$ with $f(U \setminus \{\bar{x}\}) \subseteq V$. Define the auxiliary function $\tilde{f}: X \rightarrow Y$ by letting $\tilde{f}(x) = f(x)$ if $x \neq \bar{x}$ and $\tilde{f}(\bar{x}) = \bar{y}$. Show that $\lim_{x \rightarrow \bar{x}} f(x) = \bar{y}$ if and only if $\tilde{f}$ is continuous at $\bar{x}$.

Solution: Let $f: X \rightarrow Y$ be a function between topological spaces as above with $\bar{x} \in X$ and $\bar{y} \in Y$. Define $\tilde{f}: X \rightarrow Y$ by $\tilde{f}(x) = f(x)$ for $x \neq \bar{x}$ and $\tilde{f}(\bar{x}) = \bar{y}$.

Suppose that $\lim_{x \rightarrow \bar{x}} f(x) = \bar{y}$. Then, for any $V \in \mathcal{O}_Y(\bar{y})$, there is some $U \in \mathcal{O}_X(\bar{x})$ such that $f(U \setminus \{\bar{x}\}) \subseteq V$. Clearly $\{\bar{y}\} \subseteq V$, so by definition of $\tilde{f}$, $\tilde{f}(U) = \tilde{f}(U \setminus \{\bar{x}\}) \cup \tilde{f}(\{\bar{x}\}) = f(U \setminus \{\bar{x}\}) \cup \{\bar{y}\} \subseteq V$. Thus, $\tilde{f}$ is continuous at $\bar{x}$.

On the other hand, if we suppose that $\tilde{f}$ is continuous at $\bar{x}$, then for any $V \in \mathcal{O}_Y(\tilde{f}(\bar{x})) = \mathcal{O}_Y(\bar{y})$, there is some $U \in \mathcal{O}_X(\bar{x})$ such that $\tilde{f}(U) \subseteq V$. By definition of $\tilde{f}$, $\tilde{f}(U \setminus \{\bar{x}\}) = f(U \setminus \{\bar{x}\})$, so we have that $f(U \setminus \{\bar{x}\}) \subseteq \tilde{f}(U) \subseteq V$. Thus, $\lim_{x \rightarrow \bar{x}} f(x) = \bar{y}$.

Problem (Problem I.5): Show that $\lim_{x \rightarrow \bar{x}} f(x) = \bar{y}$ if and only if for all nets $(x_i)_i$ in $X \setminus \{\bar{x}\}$ that converge to $\bar{x}$, we have that $(f(x_i))_i$ converges to $\bar{y}$.

Solution: Let $f: X \rightarrow Y$ and $\tilde{f}: X \rightarrow Y$ be functions between topological spaces as defined previously with $\bar{x} \in X$ and $\bar{y} \in Y$. We know from II.4 that $\lim_{x \rightarrow \bar{x}} f(x) = \bar{y}$ if and only if $\tilde{f}$ is continuous at $\bar{x}$. However, we also know that the continuity of $\tilde{f}$ at $\bar{x}$ is true if and only if for any net $(x_i)_{i \in I}$ in X with $x_i \rightarrow \bar{x}$, then $\tilde{f}(x_i) \rightarrow \tilde{f}(\bar{x}) = \bar{y}$.

In other words, to prove the desired statement we only need to show that for any net $(x_i)_{i \in I}$ in X with $x_i \rightarrow \bar{x}$ we have $\tilde{f}(x_i) \rightarrow \bar{y}$ if and only if for any net $(x_i)_{i \in I}$ in $X \setminus \{\bar{x}\}$ with $x_i \rightarrow \bar{x}$ we have $f(x_i) \rightarrow \bar{y}$.

The forward direction is clear because if any convergent net in X implies that the image of the net under $\tilde{f}$ converges, then for any net $(x_i)_{i \in I}$ in $X \setminus \{\bar{x}\}$ (which is still a net in X) with $x_i \rightarrow \bar{x}$, we must have that $\tilde{f}(x_i) \rightarrow \bar{y}$. But, since the net is in $X \setminus \{\bar{x}\}$ we also have that $\tilde{f}(x_i) = f(x_i)$ for any $i \in I$, so in fact $f(x_i) \rightarrow \bar{y}$ as desired.

Suppose on the other hand that we know any convergent net in $X \setminus \{\bar{x}\}$ implies the image of the net under f converges and we are given some net $(x_i)_{i \in I}$ in X with $x_i \rightarrow \bar{x}$. Let $J = \{i \in I : x_i \neq \bar{x}\}$.

Suppose that J is cofinal. As $x_i \rightarrow \bar{x}$, for any $U \in \mathcal{O}_X(\bar{x})$, there is some $u \in I$ such that if $i \geq u$, then $x_i \in U$. As J is cofinal, there is some $v \in J$ such that $v \geq u$, so for any $j \in J$ with $j \geq v$, we have $x_j \in U$. In other words, $(x_j)_{j \in J}$ is a net in $X \setminus \{\bar{x}\}$ converging to $\bar{x}$. By assumption, we then have that $(f(x_j))_{j \in J} \rightarrow \bar{y}$. Since $f(x_j) = \tilde{f}(x_j)$ for any $j \in J$ and $f(x_i) = \tilde{f}(x_i) = \bar{y}$ for any $i \in I \setminus J$, we then have that

$$(\tilde{f}(x_i))_{i \in I} \rightarrow \bar{y}.$$

If J is not cofinal, then there is some $N \in I$ such that for any $j \in J$, $j \leq N$. It is clear that $(x_i)_{N \leq i \in I}$ is then a net in $X \setminus \{\bar{x}\}$ converging to $\bar{x}$. So, $(f(x_i))_{N \leq i \in I} \rightarrow \bar{y}$ and again we conclude that $(\tilde{f}(x_i))_{i \in I} \rightarrow \bar{y}$.

Problem (Problem I.6): Let Y be a subset of a topological space $(X, \mathcal{T}_X)$. Recall that closure is defined by $\bar{Y} = \bigcap \{F \in \mathcal{P}(X) : F \text{ is closed and } Y \subseteq F\}$. Show that

$$\bar{Y} = \{x \in X : \text{for all } U \in \mathcal{O}_X(x), U \cap Y \neq \emptyset\}.$$

Solution: Let Y be a subset of a topological space $(X, \mathcal{T}_X)$ and $\bar{Y} := \bigcap_{F \in \Gamma} F$ where

$$\Gamma = \{F \in \mathcal{P}(X) : F^c \in \mathcal{T}_X \text{ and } Y \subseteq F\}.$$

Furthermore, denote $K = \{x \in X : \text{for all } U \in \mathcal{O}_X(x), U \cap Y \neq \emptyset\}$.

Suppose that $x \in K$ and $F \in \Gamma$. Suppose towards a contradiction that $x \notin F$ so that $x \in F^c$. As $F \in \Gamma$, $F^c \in \mathcal{T}_X$, so it is a neighborhood of x , so $F^c \cap Y \neq \emptyset$ because $x \in K$. However, as $F \in \Gamma$, $Y \subseteq F \Rightarrow Y \cap F^c = \emptyset$, which is a contradiction. Thus, we must have that $x \in F$ for any $F \in \Gamma$, so $x \in \bar{Y}$. Thus, $K \subseteq \bar{Y}$.

Suppose on the other hand that $x \in \bar{Y}$ so that $x \in F$ for any $F \in \Gamma$. Suppose towards a contradiction that there is some $U \in \mathcal{O}_X(x)$ such that $U \cap Y = \emptyset$. Then, we have that $Y \subseteq U^c$ and $(U^c)^c \in \mathcal{T}_X$, so $U^c \in \Gamma$, but $x \notin U^c$ which contradicts that $x \in \bar{Y}$.

Thus, for any $U \in \mathcal{O}_X(x)$, we must have that $U \cap Y \neq \emptyset$, so $x \in K$ and $\bar{Y} \subseteq K$.

Therefore,

$$\bar{Y} = \{x \in X : \text{for all } U \in \mathcal{O}_X(x), U \cap Y \neq \emptyset\}.$$

Problem (Problem I.7): Show that $x \in \bar{Y}$ if and only if there exists a net $(y_i)_i$ contained in Y such that $\lim_i y_i = x$.

Solution: Let Y be a subset of a topological space $(X, \mathcal{T}_X)$.

Suppose that there exists some net $(y_i)_{i \in I}$ in Y with $\lim_i y_i = x$. Then, for all $U \in \mathcal{O}_X(x)$, there is some $N \in$

I such that for all $i \geq N$, $y_i \in U$. Thus, $y_N \in U$, so that $U \cap Y \neq \emptyset$. Thus, $x \in \overline{Y}$.

Now, suppose otherwise that $x \in \overline{Y}$. If $x \in Y$, then clearly the sequence $(y_i)_{i \in \mathbb{N}}$ where $y_i = x$ for all $i \in \mathbb{N}$ is a net in Y with $\lim_i y_i = x$, so suppose that $x \notin Y$.

The function $\chi: X \rightarrow \{0, 1\}$ given by $\chi(u) = \mathbb{1}\{u \in Y\}$ is then not continuous at x because $\{0\}$ is an open neighborhood of $\chi(x) = 0$. Yet, for any $U \in \mathcal{O}_X(x)$, as $x \in \overline{Y}$, $U \cap Y \neq \emptyset$, so $\chi(U) = \{0, 1\} \not\subseteq \{0\}$. Let $(I, \leq)$ be the directed set given by

$$I = \{(U, z) \in \mathcal{O}_X(x) \times X : z \in U\}$$

with the relation

$$(U_1, x_1) \leq (U_2, x_2) \text{ if and only if } U_2 \subseteq U_1$$

and define the net from I to X by $(U, z) \mapsto z$.

By construction, the above net converges to x . However, for any $(U, z) \in I$, since $U \cap Y \neq \emptyset$, as $x \in \overline{Y}$, we can take some $w \in Y$ such that $(U, w) \in I$. As such, $(U, z) \leq (U, w)$ and $\chi(w) = 1$. In other words, the set $J = \{i \in I : \chi(x_i) \neq 0\}$ is a cofinal subset of I .

Let $(y_j)_{j \in J}$ be the net on X above restricted to J . By construction, $\chi(y_j) \neq 0$ for any $j \in J$, so $(y_j)_{j \in J}$ is actually a net on Y . Furthermore, since $(x_i)_{i \in I}$ converged to x , for any $U \in \mathcal{O}_X(x)$, there is some $N \in I$ such that for all $i \geq N$, $x_i \in U$. As J is a cofinal subset of I , there is some $M \in J$ such that $N \leq M$, so for any $j \in J$ with $j \geq M$, we must have that $y_j \in U$. In other words, $\lim_j y_j = x$.

Problem (Problem I.8): Let $(x_i)_i$ be a net in a topological space $(X, \mathcal{T}_X)$ converging to some element $\bar{x}$. Let F be a closed subset such that there exists $k \in I$ such that for all $i \in I$ with $i \geq k$, $x_i \in F$. Prove that one can “pass to the limit” and affirm that $\bar{x} \in F$.

Solution: Let $(x_i)_{i \in I}$ be a net in some topological space $(X, \mathcal{T}_X)$ which converges to some element $\bar{x}$ and $F \subseteq X$ be a closed subset such that there exists $k \in I$ such that for all $i \in I$ with $i \geq k$, $x_i \in F$.

Let $J = \{i \in I : i \geq k\}$. It is clear that $(x_i)_{i \in J}$ is still a net converging to $\bar{x}$ and $(x_i)_{i \in J}$ is contained within F . Since F is closed, $F = \overline{F}$, so we obtain that $\bar{x} \in F$ by I.7.

Problem II

Problem (Problem II.1): Let V and W be $\mathbb{K}$ -TVSs, and let $T: V \rightarrow W$ be a continuous linear map. Show that if $(x_i)_i$ is a Cauchy net in V , then $(Tx_i)_i$ is a Cauchy net in W .

Solution: Let U be a neighborhood of 0 in W . Then, the inverse image $T^{-1}(U)$ is an open neighborhood of 0 in V . There is some index k such that for all $i, j \in I$ with $i, j \geq k$, we have that $x_i - x_j \in T^{-1}(U)$. Yet, this means that $Tx_i - Tx_j \in U$ for all $i, j \in I$ with $i, j \geq k$. This gives the desired result.

Problem (Problem II.2): Let $(V, \mathcal{T}(\mathcal{N}))$ be a LCTVS, and let $(x_i)_i$ be a net in V . Let $\bar{x} \in V$ be some element. Show that the following are equivalent:

- (i) the net $(x_i)_i$ converges to $\bar{x}$;
- (ii) for all $\rho \in \mathcal{N}$ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $\rho(x_i - \bar{x}) \leq \varepsilon$;
- (iii) for all continuous seminorms ρ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $\rho(x_i - \bar{x}) \leq \varepsilon$.

Solution: Since translation by a vector is a homeomorphism, we will assume that $\bar{x} = 0$ and prove the results accordingly.

Suppose $(x_i)_i \rightarrow 0$. If ρ is a defining seminorm, then we observe that the subbasic open set

$$U(0, \rho, \varepsilon) = \{y \in V : \rho(y) < \varepsilon\}$$

is an open neighborhood of 0. In particular, this means there is some $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $x_i \in U(0, \rho, \varepsilon)$, or that $\rho(x_i) < \varepsilon$. This gives (ii).

To show (iii) from (ii), one observes that if ρ is a continuous seminorm, then there are defining seminorms $\rho_1, \dots, \rho_n$ and a constant C such that

$$\rho(x) \leq C \max\{\rho_1(x), \dots, \rho_n(x)\}.$$

Given $\varepsilon > 0$, for each $j = 1, \dots, n$ we select $k_j \in I$ such that for all $i_j \in I$ satisfying $i_j \geq k_j$, one has $\rho(x_{i_j}) < \varepsilon/C$. There is some k such that $k \geq k_j$ for each $j = 1, \dots, n$. Then, for any $i \in I$ with $i \geq k$, one has $\rho(x_i) \leq \varepsilon$, yielding (iii).

Finally, assuming (iii) holds, we observe that for any neighborhood U of 0, there is some continuous seminorm ρ and some $\varepsilon > 0$ such that

$$U(0, \rho, \varepsilon) = \{y \in V : \rho(y) < \varepsilon\} \subseteq U.$$

In particular, this means that there exists some $k \in I$ such that for all $i \in I$ with $i \geq k$, one has $\rho(x_i) \leq \varepsilon/2 < \varepsilon$ by the assumption of (iii). In particular, this means for all $i \geq k$, one has $x_i \in U(0, \rho, \varepsilon) \subseteq U$. Thus, the net $(x_i)_i$ converges to 0.

Problem (Problem II.3): Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space, and let $(x_i)_i$ be a net in V . Show that the following are equivalent:

- (i) the net $(x_i)_i$ is Cauchy;
- (ii) for all $\rho \in \mathcal{N}$ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i, j \in I$ with $i, j \geq k$, one has $\rho(x_i - x_j) \leq \varepsilon$;
- (iii) for all continuous seminorms ρ and for all $\varepsilon > 0$, there exists $k \in I$ such that for all $i, j \in I$ with $i, j \geq k$, one has $\rho(x_i - x_j) \leq \varepsilon$.

Solution: Suppose $(x_i)_i$ is Cauchy. Let ρ be a defining seminorm for the topology on V . Then, the subbasic open set

$$U(0, \rho, \varepsilon) = \{y \in V : \rho(y) < \varepsilon\}$$

is an open neighborhood of 0. Therefore, there is k such that for all $i, j \geq k$, one has $x_i - x_j \in U(0, \rho, \varepsilon)$. Equivalently, we have that $\rho(x_i - x_j) \leq \varepsilon$, giving (ii).

Now, suppose (ii) holds. Then, if ρ is a continuous seminorm, there are defining $\rho_1, \dots, \rho_n$ and there is a constant C such that for all $x \in V$, one has

$$\rho(x) \leq C \max\{\rho_j(x) : j = 1, \dots, n\}.$$

Since (ii) holds, it follows that there is some k such that for all $i, j \geq k$, one has $\rho_1(x_i - x_j), \dots, \rho_n(x_i - x_j) \leq \varepsilon/C$, whence $\rho(x_i - x_j) \leq \varepsilon$.

Finally, suppose (iii) holds. Let U be an open neighborhood of 0. Then, there is some continuous seminorm ρ and some $\varepsilon > 0$ such that $U(0, \rho, \varepsilon) \subseteq U$. It follows that there exists $k \in I$ such that $\rho(x_i - x_j) < \varepsilon$ for all $i, j \geq k$. It follows that $x_i - x_j \in U(0, \rho, \varepsilon) \subseteq U$, so that $(x_i)_i$ is Cauchy.

Problem (Problem II.4): Let V be a Banach space. Show that this space satisfies the new definition of complete.

Solution: Suppose $(x_i)_i$ is a Cauchy net. We will assume without loss of generality that the directed set I is infinite. Notice that Banach spaces are first countable. That is, for any neighborhood U of 0, there is some natural number $n \geq 1$ such that $U(0, 1/n) \subseteq U$.

Using the countable family of neighborhoods $\{U(0, 1/n) : n \geq 1\}$, we may extract a sequence in the net as follows. Select k_1 such that for all $i, j \geq k_1$, we have $\|x_i - x_j\| \leq 1$; thus, for all $i \geq k_1$, we have $\|x_i - x_{k_1}\| \leq 1$. Since $I \setminus \{k_1\}$ is still infinite, it follows that we may find k'_2 such that for all $i \geq k_2$, $\|x_i - x_{k'_2}\| \leq \frac{1}{2}$, so if we let k_2 be an upper bound for $\{k_1, k'_2\}$, then for all $i \geq k_2$, we have that $\|x_i - x_{k_2}\| \leq 1/2$. Inductively, given $\{k_1, \dots, k_n\}$, we may find $k_{n+1} \in I$ since I is infinite such that for all $i \geq k_{n+1}$, we have $\|x_i - x_{k_{n+1}}\| \leq \frac{1}{n+1}$. This gives us a countable subnet $(x_{k_j})_{j \in \mathbb{N}}$ that satisfies

$$\|x_{k_{n+1}} - x_{k_n}\| \leq \frac{1}{n},$$

which yields that this countable subnet (i.e., sequence) is Cauchy. Since any Cauchy net contains a countable Cauchy subnet (i.e., a Cauchy sequence), it follows that any Cauchy net converges.

Problem (Problem II.5): Show that a TVS is a Fréchet space if and only if it is locally convex, metrizable, and complete.

Solution: We will show that if V is locally convex and metrizable, with countable family of defining seminorms $\{\rho_k : k \in \mathbb{N}\}$ and distance metric given by

$$d(x, y) = \sum_{j=0}^{\infty} 2^{-j} \min(1, \rho_j(x - y))$$

that induces the topology, then a net $(x_i)_i$ is d -Cauchy if and only if it is Cauchy as a net.

Suppose $(x_i)_i$ is d -Cauchy. Let ρ_j be a defining seminorm. Observe that

$$\min(1, \rho_j(x - y)) \leq 2^j d(x - y).$$

Let $\varepsilon > 0$. Then, since $(x_i)_i$ is d -Cauchy, there is some k such that $2^j d(x_i - x_j) < \min(1, \varepsilon)$ for all $u, v \geq k$. Therefore, since

$$\begin{aligned} \min(1, \rho_j(x_u - x_v)) &\leq 2^j d(x_u - x_v) \\ &< \min(1, \varepsilon), \end{aligned}$$

it follows that $\rho_j(x_u - x_v) < \min(1, \varepsilon) \leq \varepsilon$ for all $u, v \geq k$, so that $(x_i)_i$ is Cauchy as a net.

Now, suppose $(x_i)_i$ is Cauchy as a net. Let $\varepsilon > 0$, and let K be sufficiently large such that $\sum_{j=K}^{\infty} 2^{-j} < \varepsilon/2$. Notice that

$$\rho = \sum_{j=0}^{K-1} 2^{-j} \rho_j$$

is a continuous seminorm. Therefore, there is some k_0 such that for all $u, v \geq k_0$, we have $\rho(x_u - x_v) < \varepsilon/2$. This gives, for all $u, v \geq k_0$,

$$\begin{aligned} d(x_u, x_v) &= \sum_{j=0}^{K-1} 2^{-j} \min(1, \rho_j(x_u - x_v)) + \sum_{j=K}^{\infty} 2^{-j} \min(1, \rho_j(x_u - x_v)) \\ &\leq \sum_{j=0}^{K-1} 2^{-j} \rho_j(x_u - x_v) + \sum_{j=K}^{\infty} 2^{-j} \\ &< \varepsilon. \end{aligned}$$

Therefore, $(x_i)_i$ is d -Cauchy.

In particular, this gives that a space is Fréchet if and only if d -Cauchy nets converge if and only if Cauchy nets converge if and only if the space is complete as a topological vector space.

Problem (Problem II.6): For any $\omega \in [0, \infty)^{\mathbb{N}}$ and $x = (x_n)_n \in s_0(\mathbb{N}, \mathbb{K})$, we let $\|x\|_\omega = \sup_{n \geq 0} \omega_n |x_n|$. Show that these seminorms define the topology of $s_0(\mathbb{N}, \mathbb{K})$.

Solution: Let $\mathcal{T}_1$ be the original topology generated by all the seminorms of $s_0(\mathbb{N}, \mathbb{K})$, and let $\mathcal{T}_2$ be this new topology. If $(x_i)_i$ converges to 0 in $\mathcal{T}_1$, then since the seminorms $\|x\|_\omega$ as defined in this question are seminorms on $s_0(\mathbb{N}, \mathbb{K})$, it follows that they must converge in $\mathcal{T}_2$.

In the reverse direction, notice that for any $\rho \in \text{SN}(s_0(\mathbb{N}, \mathbb{K}))$, we consider the sequence(s) given by $\omega^{(n)} = \rho(e_n)e_n$. Then, it follows that for all $x \in X$, we have

$$\begin{aligned} \rho(x) &\leq \sum_{n=0}^{\infty} \rho(e_n) |x_n| \\ &= \sum_{n=0}^{\infty} \left(\sup_{j \in \mathbb{N}} \omega_j^{(n)} |x_n| \right). \end{aligned}$$

The latter gives the desired upper bound that establishes the continuity of any such ρ .

Problem (Problem II.7): Let $y = (y_n)_n \in \mathbb{K}^{\mathbb{N}}$ such that for all $(\lambda_n)_n \in (0, \infty)^{\mathbb{N}}$, there exists $N \geq 0$ such that for all $n \geq N$, $|y_n| \leq \lambda_n$. Show that y is finitely supported.

Solution: Suppose y were not finitely supported. Then, there is a subsequence $(y_{n_k})_k$ such that $(|y_{n_k}|)_k \in (0, \infty)^{\mathbb{N}}$. Consider now the sequence $(\lambda_n)_n$ defined by

$$\lambda_n = \begin{cases} \frac{1}{n+1} & n \neq n_k \text{ for any } k \\ \frac{1}{2} |y_{n_k}| & n = n_k. \end{cases}$$

For any $N \in \mathbb{N}$, there exists some $n_k \geq N$ since $(n_k)_k$ is an unbounded sequence, so that $\lambda_{n_k} < |y_{n_k}|$. Thus, any such infinitely supported sequence y cannot satisfy the hypothesis.

Problem (Problem II.8): Let $(x_i)_i$ be a net in $s_0(\mathbb{N}, \mathbb{K})$. We will write $x_i = (x_i(n))_{n \geq 0}$ for each $i \in I$. Suppose $(x_i)_i$ is Cauchy in $s_0(\mathbb{N}, \mathbb{K})$. Show that there is a sequence $y = (y_n)_n \in \mathbb{K}^{\mathbb{N}}$ such that for all $n \geq 0$, the net $(x_i(n))_i$ converges to y_n in $\mathbb{K}$.

Solution: The seminorms $\rho_n(x_i) = |x_i(n)|$ are continuous in $s_0(\mathbb{N}, \mathbb{K})$. It follows that for each n , we have that the nets $(x_i(n))_{i \in I}$ are Cauchy in $\mathbb{K}$, so that they converge to $y(n)$. That is, we may define $(y(n))_n$ by taking pointwise limits.

We claim that $(x_i)_i \rightarrow y$ in $s_0(\mathbb{N}, \mathbb{K})$. Let $\omega \in [0, \infty)^{\mathbb{N}}$. For each n , we have some k such that for all $j \geq k$, we have $|x_j(n) - y(n)| < \varepsilon/2\omega(n)$. It follows that for each n , we have $\sup_{j \geq k} \omega(n) |x_j(n) - y(n)| < \varepsilon/2$, whence

$$\sup_{n \in \mathbb{N}} \sup_{j \geq k} \omega(n) |x_j(n) - y(n)| \leq \varepsilon/2,$$

so that for all $j \geq k$, we have that $\sup_{n \in \mathbb{N}} \omega(n) |x_j(n) - y(n)| < \varepsilon$. Thus, $(x_i)_i \rightarrow y$ in $s_0(\mathbb{N}, \mathbb{K})$.

Problem (Problem II.9): With the same hypotheses as in the previous question, show that $y \in s_0(\mathbb{N}, \mathbb{K})$ and $(x_i)_i \rightarrow y$ in the topology of $s_0(\mathbb{N}, \mathbb{K})$.

Solution: Rewriting the Cauchy hypothesis, we have that for all $\omega \in [0, \infty)^{\mathbb{N}}$ and for all $\varepsilon > 0$, there exists some $k \in I$ such that for all $i, j \geq k$ and for all $n \geq 0$, we have that $|x_i(n) - x_j(n)| \leq \frac{\varepsilon}{\omega_n}$. (For the sake of argument, we will always assume that $\omega_n > 0$.) Having defined $y(n)$ as the pointwise limit of $(x_i(n))_i$, we may take the limit with respect to j , yielding that for all $n \geq 0$ and for all $i \geq k$, we have

$$|x_i(n) - y(n)| \leq \frac{\varepsilon}{\omega_n}.$$

Now, observe that for all $i \geq k$, we have

$$|y(n)| \leq |x_i(n) - y(n)| + |x_i(n)|$$

$$\leq \frac{\varepsilon}{\omega_n} + |x_i(n)|.$$

Let N be the first index for which $x_i(n) = 0$ for all $n \geq N$. Then, we have that $|y(n)| \leq \frac{\varepsilon}{\omega_n}$. Letting $\lambda_n = \frac{\varepsilon}{\omega_n}$, it follows that, since ω_n is any (nonzero) sequence, it follows that $(y(n))_n$ satisfies the hypotheses of II.7, so that $y \in s_0(\mathbb{N}, \mathbb{K})$. Since, for all $\omega \in [0, \infty)^\mathbb{N}$ and for all $n \geq 0$, we have that $\omega_n |x_i(n) - y(n)| \leq \varepsilon$ whenever $i \geq k$, it follows that $(x_i)_i \rightarrow y$ in $s_0(\mathbb{N}, \mathbb{K})$.

Problem III

Problem (Problem III.1): We say a neighborhood U of the origin in a topological vector space (V, T) is balanced if and only if for all $x \in U$ and all $\lambda \in \mathbb{K}$ with $|\lambda| \leq 1$, we have $\lambda x \in U$. Show that for all open neighborhoods $U \in \mathcal{O}_V(0)$, there exists $B \in \mathcal{O}_V(0)$ with $B \subseteq U$ and B balanced.

Solution: Let U be an open neighborhood of 0. Then, since the scalar multiplication map $m: \mathbb{K} \times V \rightarrow V$ is continuous, it follows that $m^{-1}(U)$ is an open subset of $\mathbb{K} \times V$. Therefore, there is a basic open subset $U(0, \varepsilon) \times W \subseteq m^{-1}(U)$, where

$$U(0, \varepsilon) = \{z \in \mathbb{K} : |z| < \varepsilon\}.$$

It follows that

$$m(U(0, \varepsilon) \times W) = \bigcup_{|z| < \varepsilon} zW$$

is a balanced open subset of U containing 0. This yields our desired result.

Problem (Problem III.2): Show that, in a topological vector space, if two subsets A and B are bounded, then $A \cup B$ is bounded.

Solution: Let $U \in \mathcal{O}_V(0)$ be a neighborhood of 0. Then, there is a balanced subset $W \in \mathcal{O}_V(0)$ with $W + W \subseteq U$, where $W + W$ denotes the Minkowski sum of W with itself. Such a neighborhood can be found by obtaining a balanced subset W' of U , then taking $W = \frac{1}{2}W'$.

There is ε_1 such that $\varepsilon_1 A \subseteq W$ and $\varepsilon_2 B \subseteq W$. Taking $\varepsilon = \min(\varepsilon_1, \varepsilon_2)$, we observe that $\frac{\varepsilon}{\varepsilon_1} \leq 1$ and $\frac{\varepsilon}{\varepsilon_2} \leq 1$. Therefore, $\varepsilon A \subseteq \frac{\varepsilon}{\varepsilon_1} W \subseteq W$ since W is balanced. Similarly, $\varepsilon B \subseteq \frac{\varepsilon}{\varepsilon_2} W \subseteq W$. Therefore, $\varepsilon(A \cup B) = \varepsilon A \cup \varepsilon B \subseteq W \cup W = W$.

Problem (Problem III.3): We define sets of nonnegative sequences with specified behavior at infinity as follows

$$\begin{aligned} s_+(\mathbb{N}, \mathbb{K}) &= s(\mathbb{N}, \mathbb{K}) \cap [0, \infty)^\mathbb{N} \\ s_{0,+}(\mathbb{N}, \mathbb{K}) &= s_0(\mathbb{N}, \mathbb{K}) \cap [0, \infty)^\mathbb{N}. \end{aligned}$$

Show that $A \subseteq s(\mathbb{N}, \mathbb{K})$ is bounded if and only if there exists $\omega \in s_+(\mathbb{N}, \mathbb{K})$ such that for all $x = (x_n)_n \in A$ and for all $n \geq 0$, we have $|x_n| \leq \omega_n$. Show that a similar result holds for the finitely supported sequences.

Solution: The solution to this problem is detailed at the beginning of the next problem.

Problem IV

Throughout the proof, write $s_0 = s_0(\mathbb{N}, \mathbb{K})$, $s = s(\mathbb{N}, \mathbb{K})$, and $s' = s'(\mathbb{N}, \mathbb{K})$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

For these spaces, we add a $+$ in the subscript when we intersecting them with $[0, \infty)^\mathbb{N}$ and call them the positive part.

Recall that $\langle n \rangle = (1 + n^2)^{1/2}$ and

$$\|x\|_k = \sup_{n \geq 0} \langle n \rangle^k |x_n|$$

$$\|x\|_\omega = \sup_{n \geq 0} \omega_n |x_n|.$$

The topology of s is generated by $\|\cdot\|_k$.

By Problem II.6, the topology of s_0 is equivalently generated by $\|\cdot\|_\omega$, with $\omega \in [0, \infty)^\mathbb{N}$.

On s' , the defining weights are $\omega \in s_+$; on $\mathbb{K}^\mathbb{N}$, the defining seminorms are $x \mapsto |x_n|$. All continuous duals carry the strong topology, generated by

$$\|L\|_B = \sup_{y \in B} |L(y)|,$$

where B ranges over the nonempty bounded subsets of the original space.

We first characterize bounded sets in the four sequence spaces above. We claim that for each of the four spaces, a nonempty subset B is bounded if and only if there is a nonnegative sequence $b = (b_n)_{n \geq 0}$ such that $|x_n| \leq b_n$ for all $x \in B$ and all n , with b belonging to their positive part.

We prove all four statements directly from the fact that boundedness is equivalent to uniform boundedness of every defining seminorm on B .

First, sets dominated by a sequence b is obviously bounded since their defining seminorms are bounded by those of b .

Conversely, suppose that B is bounded and set $b_n = \sup_{x \in B} |x_n|$.

Coordinate projections are continuous in all four spaces, so each b_n is finite. This proves the assertion for $\mathbb{K}^\mathbb{N}$.

If $B \subseteq s$, interchanging the two suprema gives

$$\begin{aligned} \|b\|_k &= \sup_{x \in B} \|x\|_k \\ &< \infty \end{aligned}$$

for any $k \geq 0$, so $b \in s_+$.

If $B \subseteq s_0$ and b had infinite support, we could choose strictly increasing indices n_j and $x^{(j)} \in B$, for $j \geq 1$, with $x_{n_j}^{(j)} \neq 0$. Define $\omega_{n_j} = j/|x_{n_j}^{(j)}|$ and $\omega_n = 0$ elsewhere. This is an admissible weight for s_0 , but $\|x^{(j)}\|_\omega \geq j$, contradicting boundedness. Thus $b \in s_{0,+}$.

Finally, if $B \subseteq s'$, then for every $\omega \in s_+$,

$$\begin{aligned} \sup_{n \geq 0} \omega_n b_n &= \sup_{x \in B} \|x\|_\omega \\ &< \infty. \end{aligned} \tag{1}$$

If b did not have polynomial growth, then $(\langle n \rangle^{-j} b_n)_n$ would be unbounded for each $j \geq 1$. We can choose strictly increasing n_j such that $b_{n_j} > j \langle n_j \rangle^j$. Set $\omega_{n_j} = \langle n_j \rangle^{-j}$ and $\omega_n = 0$ elsewhere.

For every fixed $k \geq 0$,

$$\begin{aligned} \|\omega\|_k &= \sup_{j \geq 1} \langle n_j \rangle^{k-j} \\ &< \infty. \end{aligned} \tag{2}$$

Hence $\omega \in s_+$, whereas $\omega_{n_j} b_{n_j} > j$, contradicting (1). This completes the proof of the bounded set characterizations.

We need one more result before starting the proof. For x in any of these spaces, let $x^{(N)} = \sum_{n=0}^N x_n e_n$, we assert its truncations converge to x in that space. In the product topology, each coordinate eventually agrees with that of x ; in s_0 , the whole sequence eventually equals x .

If $x \in s$, then

$$\begin{aligned} \|x - x^{(N)}\|_k &\leq \frac{\|x\|_{k+1}}{\langle N+1 \rangle} \\ &\rightarrow 0. \end{aligned}$$

If $x \in s'$, choose $|x_n| \leq C\langle n \rangle^m$. For $\omega \in s_+$,

$$\begin{aligned} \|x - x^{(N)}\|_\omega &\leq C \sup_{n>N} \omega_n \langle n \rangle^m \\ &\leq \frac{C\|\omega\|_{m+1}}{\langle N+1 \rangle} \\ &\rightarrow 0. \end{aligned}$$

In particular, s_0 is dense in all four spaces, and every continuous linear form satisfies

$$L(y) = \lim_{N \rightarrow \infty} \sum_{n=0}^N L(e_n) y_n. \quad (3)$$

Thus its coefficients $L(e_n)$ determine it uniquely.

For the converse constructions, define

$$\begin{aligned} L_a(y) &= \langle a, y \rangle \\ &= \sum_{n \geq 0} a_n y_n. \end{aligned}$$

The pairing is bilinear. Set $A = \sum_{n \geq 0} \langle n \rangle^{-2} < \infty$. The following estimates show that each L_a is a continuous linear form on the corresponding space:

- If $a \in \mathbb{K}^{\mathbb{N}}$ and $y \in s_0$, the sum is finite. With $\omega_n = 2^{n+1}|a_n|$,

$$\begin{aligned} \sum_{n \geq 0} |a_n y_n| &\leq \sum_{n \geq 0} 2^{-n-1} \|y\|_\omega \\ &= \|y\|_\omega. \end{aligned}$$

- If $a \in s_0$ and $y \in \mathbb{K}^{\mathbb{N}}$, choose N such that $a_n = 0$ for $n > N$. Then

$$\sum_{n \geq 0} |a_n y_n| \leq \left(\sum_{n=0}^N |a_n| \right) \max\{|y_n| : 0 \leq n \leq N\}.$$

- If $a \in s'$ and $y \in s$, choose $|a_n| \leq C\langle n \rangle^m$. Then

$$\begin{aligned} \sum_{n \geq 0} |a_n y_n| &\leq C \sum_{n \geq 0} \langle n \rangle^m |y_n| \\ &\leq CA \|y\|_{m+2} \end{aligned}$$

- If $a \in s$ and $y \in s'$, put $\omega_n = \langle n \rangle^2 |a_n|$. Since $\|\omega\|_k = \|a\|_{k+2} < \infty$, we have $\omega \in s_+$, and

$$\begin{aligned} \sum_{n \geq 0} |a_n y_n| &= \sum_{n \geq 0} \langle n \rangle^{-2} \omega_n |y_n| \\ &\leq A \|y\|_\omega \end{aligned}$$

These estimates prove absolute convergence and continuity of L_a in each case. Moreover, $L_a(e_n) = a_n$. Consequently, once the coefficient sequence of a continuous linear functional L is shown to belong to the indicated space, (3) gives $L = L_a$. The corresponding coefficient map is then a linear bijection with inverse $a \mapsto L_a$, and the required pairing formula follows. We will use this observation in parts 1, 2, 4, and 5. The continuity and well-definedness of these bijection and their inverse will be checked separately.

Problem (Problem IV.1): Show that the map ϕ from the strong topological dual $s_0(\mathbb{N}, \mathbb{K})'$ of the space $s_0(\mathbb{N}, \mathbb{K})$ to $\mathbb{K}^{\mathbb{N}}$ which sends a linear form L to the sequence $(L(e_n))_{n \geq 0}$ is a TVS isomorphism.

Solution: For every n , the set $\{e_n\}$ is bounded in s_0 , and

$$|\phi(L)_n| = \|L\|_{\{e_n\}}$$

Thus ϕ is continuous for the product topology. For the inverse, let $B \subseteq s_0$ be nonempty and bounded. By the bounded set characterization above, choose $b \in s_{0,+}$ such that $|y_n| \leq b_n$ for all $y \in B$, and choose N such that $b_n = 0$ for $n > N$. Then

$$\|L_a\|_B \leq \sum_{n=0}^N b_n |a_n|.$$

The last seminorm is continuous for the product topology, so ϕ^{-1} is continuous.

Problem (Problem IV.2): Show that the map ψ from the strong topological dual $(\mathbb{K}^{\mathbb{N}})'$ to $s_0(\mathbb{N}, \mathbb{K})$ which sends a linear form L to the sequence $(L(e_n))_{n \geq 0}$ is a TVS isomorphism. Also, show for all $L \in (\mathbb{K}^{\mathbb{N}})'$ and all $y \in \mathbb{K}^{\mathbb{N}}$ that $L(y) = \langle \psi(L), y \rangle$.

Solution: For $L \in (\mathbb{K}^{\mathbb{N}})'$, continuity gives $C > 0$ and $N \geq 0$ such that

$$|L(y)| \leq C \max\{|y_n| : 0 \leq n \leq N\}.$$

Thus $L(e_n) = 0$ for $n > N$, so $\psi(L) \in s_0$. For any $\omega \in [0, \infty)^{\mathbb{N}}$, the set $B_\omega = \{\omega_n e_n : n \geq 0\}$ is dominated by ω , hence bounded in $\mathbb{K}^{\mathbb{N}}$. Then

$$\begin{aligned} \|\psi(L)\|_\omega &= \sup_{n \geq 0} \omega_n |L(e_n)| \\ &= \|L\|_{B_\omega}, \end{aligned}$$

so ψ is continuous.

For the inverse, let $B \subset \mathbb{K}^{\mathbb{N}}$ be nonempty and bounded. Choose a nonnegative sequence b dominating B , and set $\omega_n = 2^{n+1} b_n$. Then

$$\begin{aligned} \|L_a\|_B &\leq \sum_{n \geq 0} b_n |a_n| \\ &\leq \sum_{n \geq 0} 2^{-n-1} \|a\|_\omega \\ &= \|a\|_\omega, \end{aligned}$$

so ψ^{-1} is continuous.

Problem (Problem IV.3): Let us introduce a new space of sequences $s'(\mathbb{N}, \mathbb{K})$, which are the sequences $x = (x_n)_{n \geq 0} \in \mathbb{K}^{\mathbb{N}}$ such that there exists $k \in \mathbb{N}$ and $C \geq 0$ such that for all $n \in \mathbb{N}$, $|x_n| \leq C \langle n \rangle^k$.

On it, we define the seminorms $\|x\|_\omega = \sup_{n \geq 0} \omega_n |x_n|$ indexed by $\omega \in s_+(\mathbb{N}, \mathbb{K})$. Show that $s'(\mathbb{N}, \mathbb{K})$ is not metrizable.

Solution: Suppose, for a contradiction, that a countable subset of the defining seminorms generates the topology of s' . Write its weights as $(\omega^{(j)})_{j \geq 1} \subseteq s_+$.

Consider all positive integer combinations of these weights as $(a^{(j)})_{j \geq 1} \subseteq s_+$. Choose strictly increasing n_j such that $\langle n_j \rangle > \|a^{(j)}\|_{j+1}$. Then

$$a_{n_j}^{(j)} \leq \|a^{(j)}\|_{j+1} \langle n_j \rangle^{-j-1}$$

$$\langle n_j \rangle^{-j}.$$

Set $\omega_{n_j} = \langle n_j \rangle^{-j}$ and $\omega_n = 0$ elsewhere. As in (2), $\omega \in s_+$. By the seminorm criterion, there are $c_i > 0$ and a finite nonempty set F such that

$$\|x\|_\omega \leq \sup_{i \in F} c_i \|x\|_{\omega^{(i)}}$$

for any $x \in s'$.

Taking $x = e_n$, we get

$$\omega_n \leq \sum_{i \in F} C_i \omega_n^{(i)}.$$

The weight on the right is one of the listed combinations, say $a^{(r)}$. But $\omega_{n_r} = \langle n_r \rangle^{-r} > a_{n_r}^{(r)}$, a contradiction. By the metrizable criterion for locally convex spaces, s' is not metrizable.

Problem (Problem IV.4): Show that the map θ from $s(\mathbb{N}, \mathbb{K})'$ to $s'(\mathbb{N}, \mathbb{K})$ which sends a linear form L to the sequence $(L(e_n))_{n \geq 0}$ is a TVS isomorphism. Also, show that for all $L \in s(\mathbb{N}, \mathbb{K})'$ and all $y \in s(\mathbb{N}, \mathbb{K})$ that $L(y) = \langle \theta(L), y \rangle$.

Solution: For $L \in (s)'$, continuity gives $C > 0$ and $m \geq 0$ such that

$$|L(y)| \leq C \|y\|_m.$$

Thus $|L(e_n)| \leq C \langle n \rangle^m$, so $\theta(L) \in s'$. For any $\omega \in s_+$, the set $B_\omega = \{\omega_n e_n : n \geq 0\}$ is dominated by ω , hence bounded in s . Then

$$\begin{aligned} \|\theta(L)\|_\omega &= \sup_{n \geq 0} \omega_n |L(e_n)| \\ &= \|L\|_{B_\omega}, \end{aligned}$$

so θ is continuous. For the inverse, let $B \subseteq s$ be nonempty and bounded. Choose $b \in s_+$ dominating B , and set $\omega_n = \langle n \rangle^2 b_n$. Since $\|\omega\|_k = \|b\|_{k+2} < \infty$, we have $\omega \in s_+$. Then

$$\begin{aligned} \|L_a\|_B &\leq \sum_{n \geq 0} |a_n| b_n \\ &= \sum_{n \geq 0} \langle n \rangle^{-2} \omega_n |a_n| \\ &\leq A \|a\|_\omega, \end{aligned}$$

so θ^{-1} is continuous.

Problem (Problem IV.5): Show that the map ν from the strong topological dual $(s'(\mathbb{N}, \mathbb{K}))'$ to $s(\mathbb{N}, \mathbb{K})$ which sends a linear form L to the sequence $(L(e_n))_{n \geq 0}$ is a TVS isomorphism. Also, show that for all $L \in (s'(\mathbb{N}, \mathbb{K}))'$ and all $y \in s'(\mathbb{N}, \mathbb{K})$ that $L(y) = \langle \nu(L), y \rangle$.

Solution: For every k , the set $B_k = \{\langle n \rangle^k e_n : n \geq 0\}$ is dominated by the polynomial growth sequence $(\langle n \rangle^k)_n$, hence bounded in s' . For $L \in (s')'$, we have

$$\begin{aligned} \|\nu(L)\|_k &= \sup_{n \geq 0} \langle n \rangle^k |L(e_n)| \\ &= \|L\|_{B_k} \\ &< \infty. \end{aligned}$$

Thus $\nu(L) \in s$, and the same equality proves that ν is continuous. For the inverse, let $B \subseteq s'$ be nonempty and bounded. By the bounded set characterization above, there are $C > 0$ and $m \geq 0$ such that $|y_n| \leq C \langle n \rangle^m$

for all $y \in B$. Then

$$\begin{aligned}\|L_a\|_B &\leq C \sum_{n \geq 0} \langle n \rangle^m |a_n| \\ &\leq CA \|a\|_{m+2}.\end{aligned}$$

so v^{-1} is continuous.

Problem V

Problem: For any $d \geq 1$, we define the following generalization of $s(\mathbb{N}, \mathbb{K})$ using multisequences (or multi-dimensional sequences). Let $s(\mathbb{N}^d, \mathbb{K})$ be the vector space of multisequences $x = (x_\alpha)_{\alpha \in \mathbb{N}^d}$ such that for all $k \geq 0$,

$$\|x\|_k = \sup_{\alpha \in \mathbb{N}^d} \langle \alpha \rangle^k |x_\alpha| < \infty.$$

Here, $\langle \alpha \rangle$ is defined using the inclusion $\mathbb{N}^d \subseteq \mathbb{R}^d$ and the bracket on $\mathbb{R}^d$. We set the locally convex topology defined by the seminorms $\|\cdot\|_k$ for each $k \in \mathbb{N}$. Show that for any $d \geq 1$, $s(\mathbb{N}^d, \mathbb{K})$ is isomorphic as a topological vector space to $s(\mathbb{N}, \mathbb{K})$.

Solution: Let $d \geq 1$ and $s(\mathbb{N}^d, \mathbb{K})$ be the vector space of multisequences $x = (x_\alpha)_{\alpha \in \mathbb{N}^d} \in \mathbb{K}^{\mathbb{N}^d}$ such that for any integer $k \geq 0$,

$$\|x\|_k := \sup_{\alpha \in \mathbb{N}^d} \langle \alpha \rangle^k |x_\alpha| < \infty$$

where $\langle \alpha \rangle$ is defined using the inclusion $\mathbb{N}^d \subseteq \mathbb{R}^d$ and the bracket of $\mathbb{R}^d$.

Define $p : \mathbb{N}^2 \rightarrow \mathbb{N}$ by $p(m, n) = \frac{(m+n)(m+n+1)}{2} + n$. For any $k \in \mathbb{N}$, choose $s \in \mathbb{N}$ such that

$$\frac{s(s+1)}{2} \leq k \leq s + \frac{s(s+1)}{2},$$

which is clearly guaranteed to exist because we can always find s such that $s(s+1)/2$ is less than k and $s + s(s+1)/2$ is at least one greater than $s(s+1)/2$. This condition is equivalent to $0 \leq k - s(s+1)/2 \leq s$. By the well-ordering principle, there exists a minimal such $s \in \mathbb{N}$ such that this is true.

If there were some other $l \in \mathbb{N}$ such that $0 \leq k - l(l+1)/2 \leq l$, then we must have that $l > s$, or that $l = s + u$ for some $0 \neq u \in \mathbb{N}$. Then, $0 \leq k - (s+u)(s+u+1)/2$, meaning $u(u+1)/2 + s \leq k - s(s+1)/2$, which contradicts that $k - s(s+1)/2 \leq s$. In other words, for each $k \in \mathbb{N}$, there is a unique $s \in \mathbb{N}$ such that $s(s+1)/2 \leq k \leq s + s(s+1)/2$.

We may then take

$$\begin{aligned}n &= k - \frac{s(s+1)}{2} \\ m &= s - n = s + \frac{s(s+1)}{2}\end{aligned}$$

(which are both positive integers according to the restrictions placed on s above) so that $p(m, n) = s(s+1)/2 + k - s(s+1)/2 = k$. Since the value of s is unique, $p^{-1}(k) = (k - s(s+1)/2, s + s(s+1)/2)$ is the well-defined inverse of p . As such, p is a bijection from $\mathbb{N}^2$ to $\mathbb{N}$.

We may then define $f : \mathbb{N}^d \rightarrow \mathbb{N}$ by

$$f(\alpha_1, \dots, \alpha_d) = p(\alpha_1, p(\alpha_2, \dots, p(\alpha_{d-1}, \alpha_d) \dots)).$$

Since p is a bijection, if we denote $p^{-1}(k) = (p_1^{-1}(k), p_2^{-1}(k))$, then f has the well-defined inverse

$$f^{-1}(k) = (p_1^{-1}(k), p_1^{-1}(p_2^{-1}(k)), \dots, p_1^{-1}(p_2^{-1}(\dots p_2^{-1}(k))), p_2^{-1}(\dots p_2^{-1}(k))).$$

In other words, for any $k \in \mathbb{N}$, $f^{-1}(k) = \alpha \in \mathbb{N}^d$ where $\alpha_i = p_1^{-1}(p_2^{-1}(\dots p_i^{-1}(k)))$ if $1 \leq i < d$ and $\alpha_d = p_2^{-1}(\dots p_2^{-1}(k))$ where in either case there are i many total nested inverse functions of p being applied to k . We then have that f is then a bijection between $\mathbb{N}^d$ and $\mathbb{N}$.

Finally, we define $F : s(\mathbb{N}^d, \mathbb{K}) \rightarrow s(\mathbb{N}, \mathbb{K})$ by

$$F((x_\alpha)_{\alpha \in \mathbb{N}^d}) = (x_{f(\alpha)})_{f(\alpha) \in \mathbb{N}}$$

with its inverse similarly defined as

$$F^{-1}((x_\beta)_{\beta \in \mathbb{N}}) = (x_{f^{-1}(\beta)})_{f^{-1}(\beta) \in \mathbb{N}^d}.$$

It is clear that F is then a bijection between $s(\mathbb{N}^d, \mathbb{K})$ and $s(\mathbb{N}, \mathbb{K})$.

For any $(x_\alpha)_{\alpha \in \mathbb{N}^d}, (y_\alpha)_{\alpha \in \mathbb{N}^d} \in \mathbb{K}^{\mathbb{N}^d}$ and $u, v \in \mathbb{K}$, we can define $z_\alpha = ux_\alpha + vy_\alpha$ so that

$$\begin{aligned} F(u(x_\alpha)_{\alpha \in \mathbb{N}^d} + v(y_\alpha)_{\alpha \in \mathbb{N}^d}) &= F((z_\alpha)_{\alpha \in \mathbb{N}^d}) \\ &= (z_{f(\alpha)})_{f(\alpha) \in \mathbb{N}} \\ &= u(x_{f(\alpha)})_{f(\alpha) \in \mathbb{N}} + v(y_{f(\alpha)})_{f(\alpha) \in \mathbb{N}} \\ &= uF((x_\alpha)_{\alpha \in \mathbb{N}^d}) + vF((y_\alpha)_{\alpha \in \mathbb{N}^d}). \end{aligned}$$

shows the map is linear.

We here note that for any $m, n \in \mathbb{N}$, $p(m, n) \geq (m+n)(m+n+1)/2$, meaning $(p(m, n))^2 \geq (m+n)^2(m+n+1)^2/4 \geq (m+n)^2 \geq m^2 + n^2$, and so $(f(\alpha_1, \dots, \alpha_d)) \geq \alpha_1^2 + \dots + \alpha_d^2$. Similarly, $p(m, n) \leq (m+n+1)^2$, so applying this bound recursively in the definition of f , we find that $(f(\alpha_1, \dots, \alpha_d)) \leq C(1 + \alpha_1^{2^d} + \dots + \alpha_d^{2^d})$ for some constant $C \in \mathbb{N}$. In other words, for any $\alpha \in \mathbb{N}^d$, we have that

$$\langle \alpha \rangle \leq \langle f(\alpha) \rangle \leq C \langle \alpha \rangle^d.$$

If $(x_\alpha)_{\alpha \in \mathbb{N}^d} \in s(\mathbb{N}^d, \mathbb{K})$, then we have that

$$\|(x_\alpha)_{\alpha \in \mathbb{N}^d}\|_k = \sup_{\alpha \in \mathbb{N}^d} \langle \alpha \rangle^k |x_\alpha| < \infty$$

for any $k \geq 0$, so

$$\begin{aligned} \|F[(x_\alpha)_{\alpha \in \mathbb{N}^d}]\|_k &= \sup_{f(\alpha) \in \mathbb{N}} \langle f(\alpha) \rangle^k |x_{f(\alpha)}| \\ &\leq \sup_{\alpha \in \mathbb{N}^d} C \langle \alpha \rangle^{dk} |x_\alpha| \\ &= C \|(x_\alpha)_{\alpha \in \mathbb{N}^d}\|_{dk} \\ &< \infty. \end{aligned}$$

Thus, F is continuous because any of the defining seminorms $\|F(\cdot)\|_k$ on $s(\mathbb{N}, \mathbb{K})$ can be bounded by the continuous seminorm $C\|\cdot\|_{dk}$ on $s(\mathbb{N}^d, \mathbb{K})$.

Similarly, for any $(y_\alpha)_{\alpha \in \mathbb{N}} \in s(\mathbb{N}, \mathbb{K})$,

$$\begin{aligned} \|(y_\alpha)_{\alpha \in \mathbb{N}}\|_k &= \sup_{\alpha \in \mathbb{N}} \langle \alpha \rangle^k |y_\alpha| \\ &< \infty \end{aligned}$$

for any $k > 0$, so

$$\|F^{-1}[(y_\alpha)_{\alpha \in \mathbb{N}}]\|_k = \sup_{f^{-1}(\alpha) \in \mathbb{N}^d} \langle f^{-1}(\alpha) \rangle^k |y_{f^{-1}(\alpha)}|$$

$$\begin{aligned}
&\leq \sup_{\alpha \in \mathbb{N}} \langle \alpha \rangle^k |y_\alpha| \\
&= \left\| (y_\alpha)_{\alpha \in \mathbb{N}} \right\|_k \\
&< \infty
\end{aligned}$$

because $\langle \beta \rangle \leq \langle f(\beta) \rangle$ for any $\beta \in \mathbb{N}^d$ gives us that $\langle f^{-1}(\alpha) \rangle \leq \langle \alpha \rangle$ for any $\alpha \in \mathbb{N}$ because f is a bijection. Thus, F^{-1} is also continuous because any of the defining seminorms $\|F^{-1}(\cdot)\|_k$ on $s(\mathbb{N}^d, \mathbb{K})$ can be bounded by the continuous seminorm $\|\cdot\|_k$ on $s(\mathbb{N}, \mathbb{K})$.

Since F is a linear homeomorphism between $s(\mathbb{N}^d, \mathbb{K})$ and $s(\mathbb{N}, \mathbb{K})$, it is a TVS isomorphism.